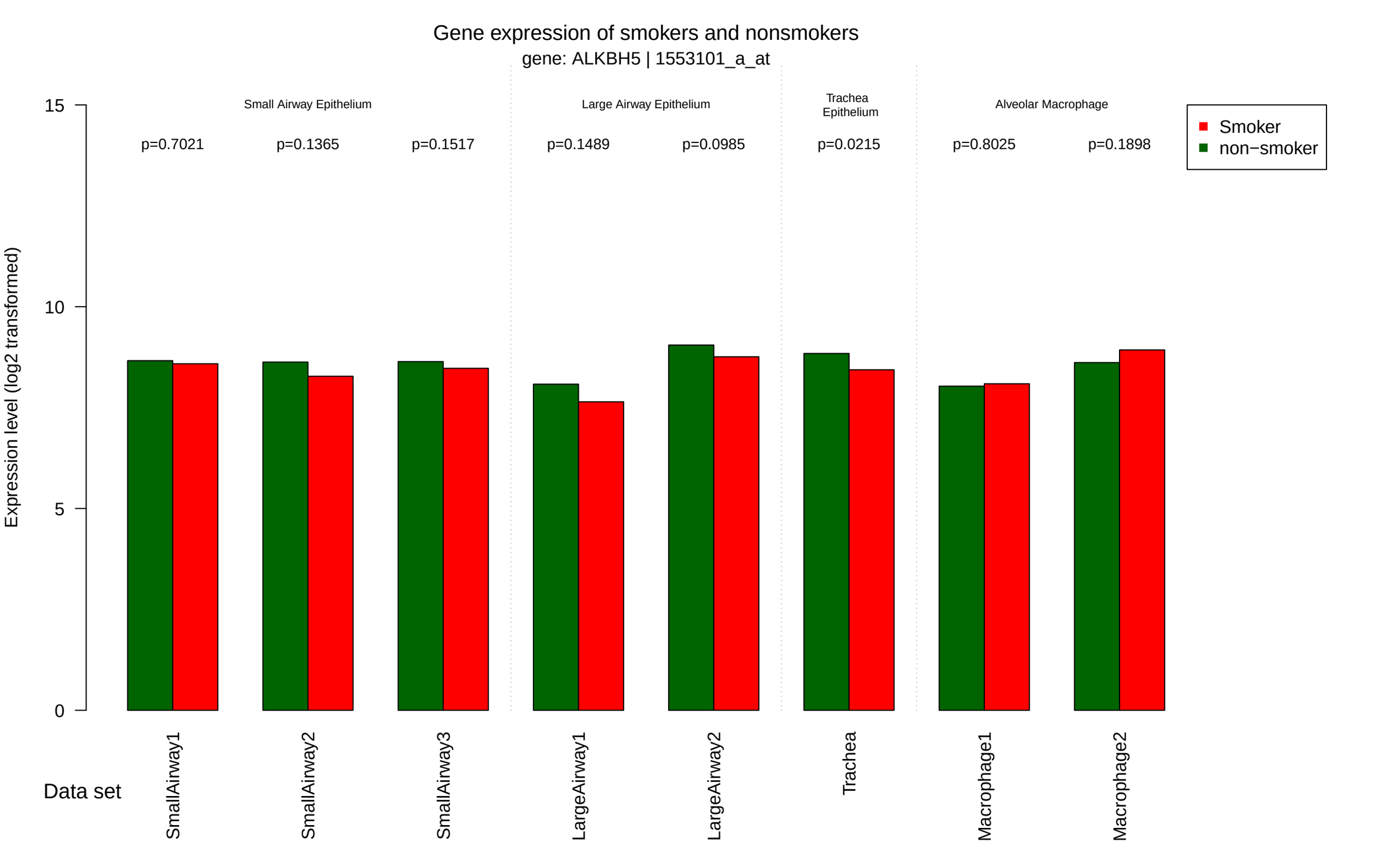
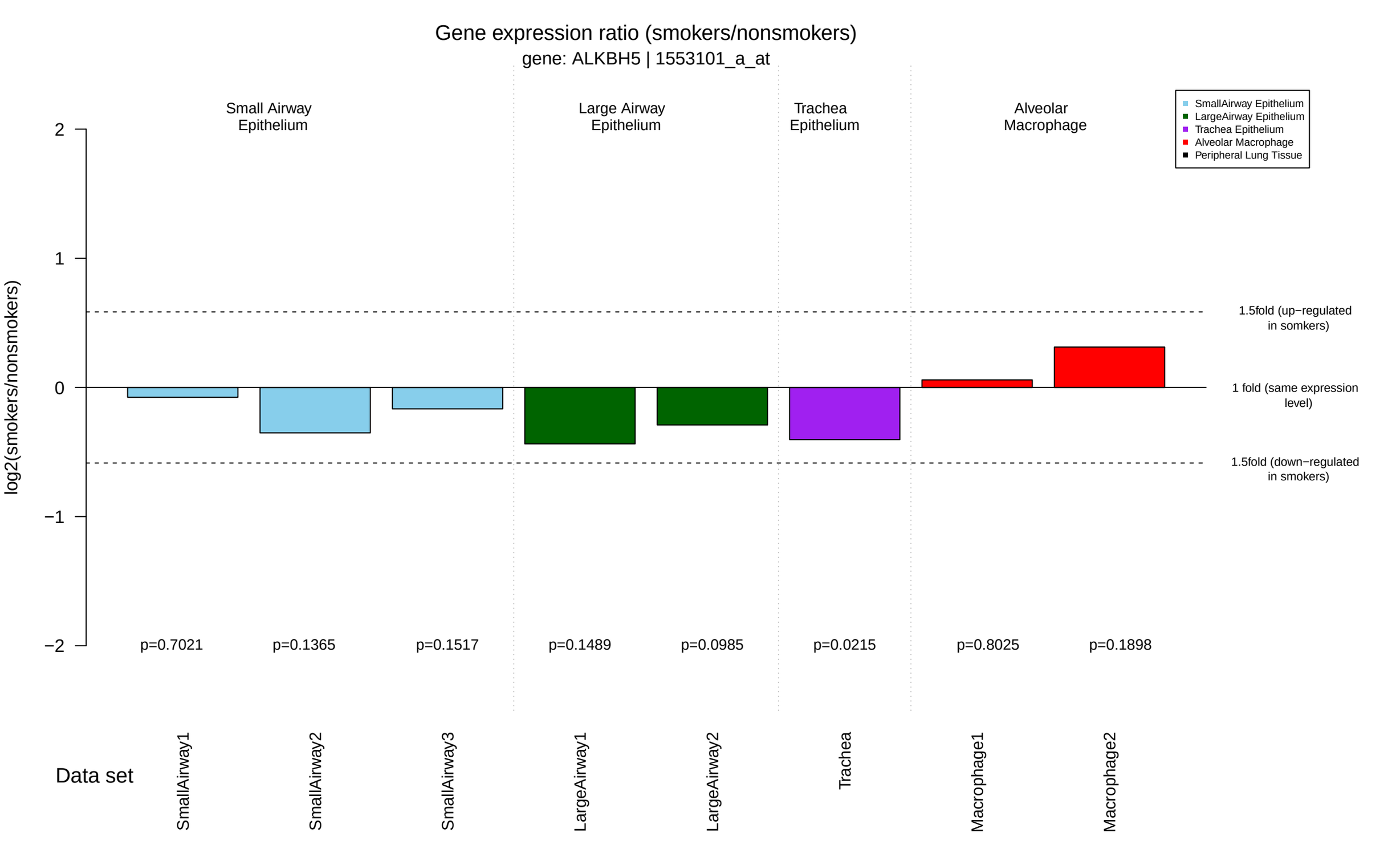
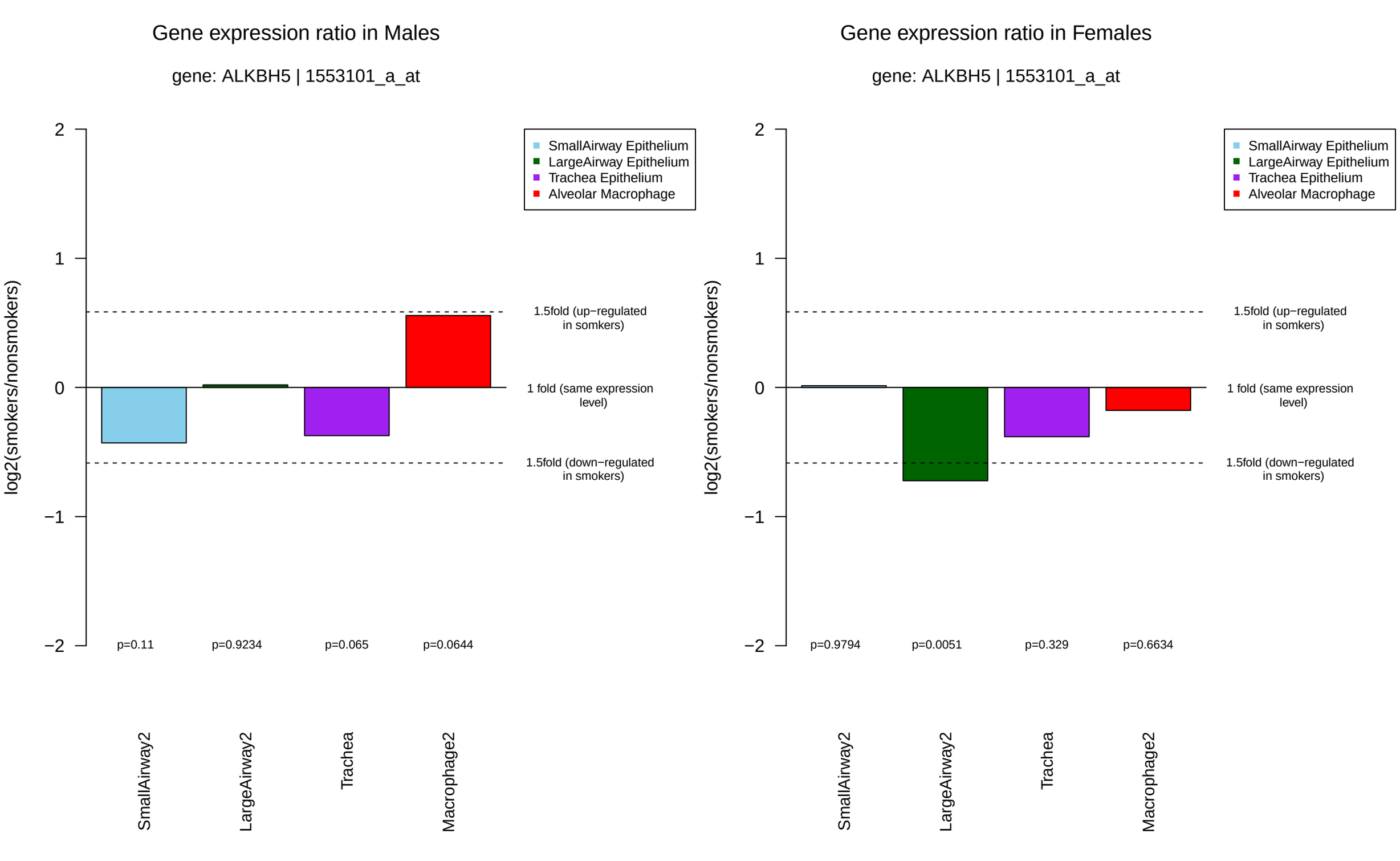
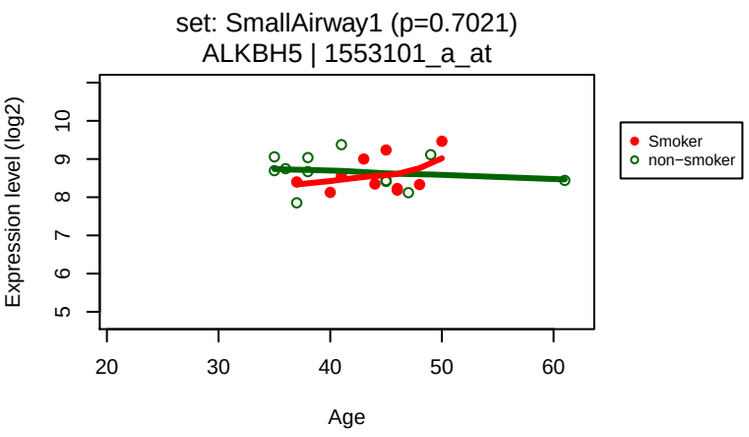
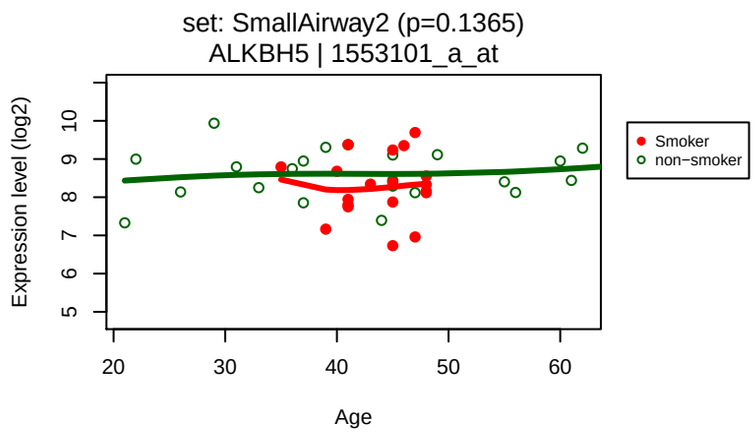
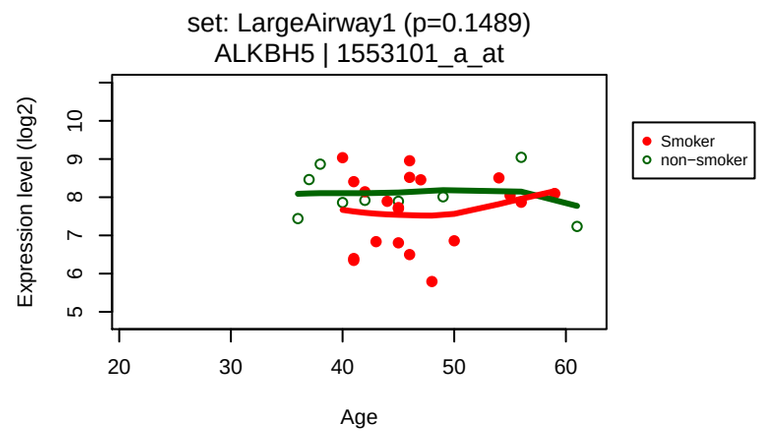

Gene expression of smokers and nonsmokers
gene: ALKBH5 | 1553101_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7021
p=0.1365
p=0.1517
p=0.1489
p=0.0985
p=0.0215
p=0.8025
p=0.1898
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

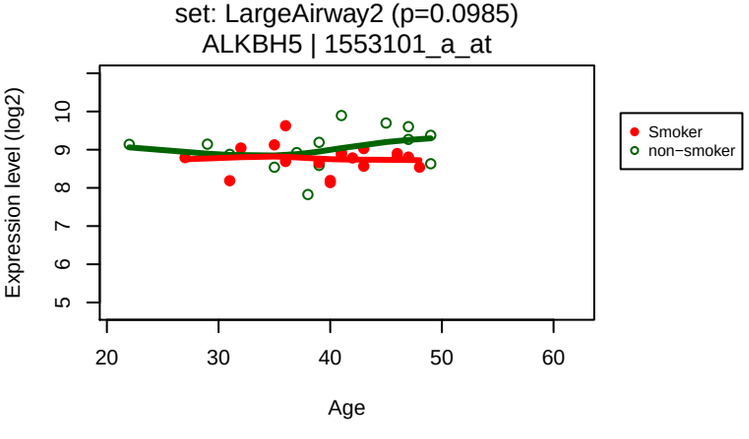

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH5 | 1553101_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7021
p=0.1365
p=0.1517
p=0.1489
p=0.0985
p=0.0215
p=0.8025
p=0.1898
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

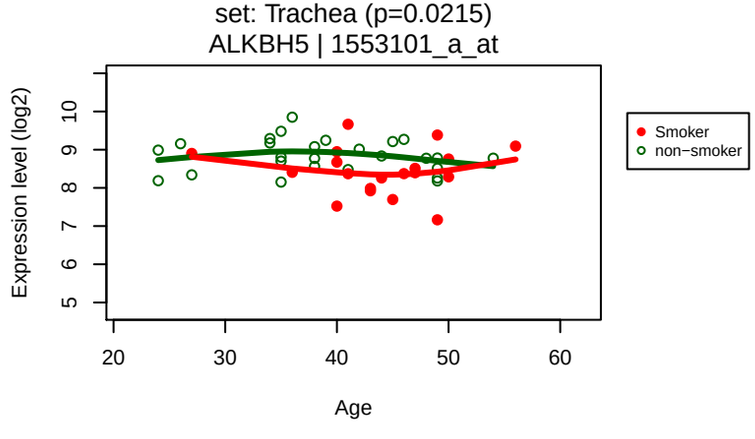

Gene expression ratio in Males
gene: ALKBH5 | 1553101_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.11
p=0.9234
p=0.065
p=0.0644
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH5 | 1553101_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9794
p=0.0051
p=0.329
p=0.6634
SmallAirway2
LargeAirway2
Trachea
Macrophage2

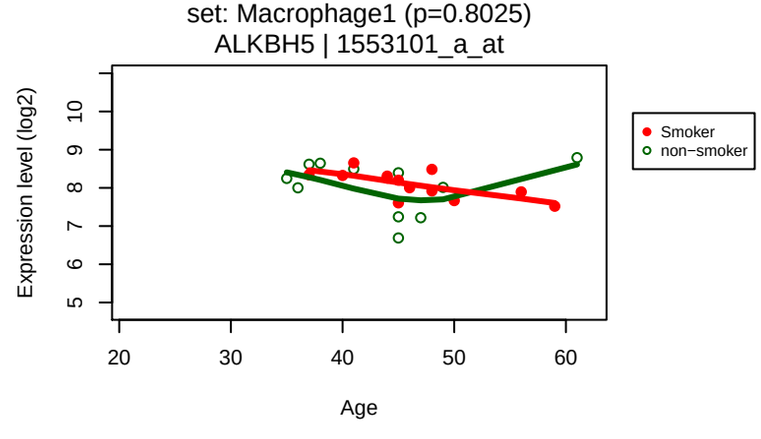

set: SmallAirway1 (p=0.7021)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker

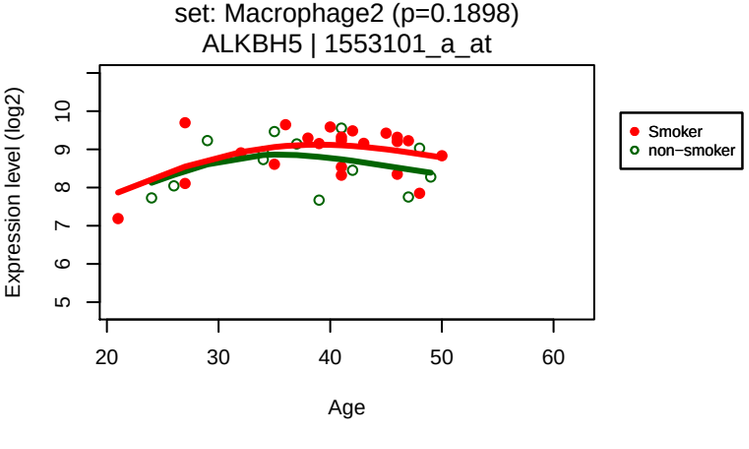

set: SmallAirway2 (p=0.1365)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker

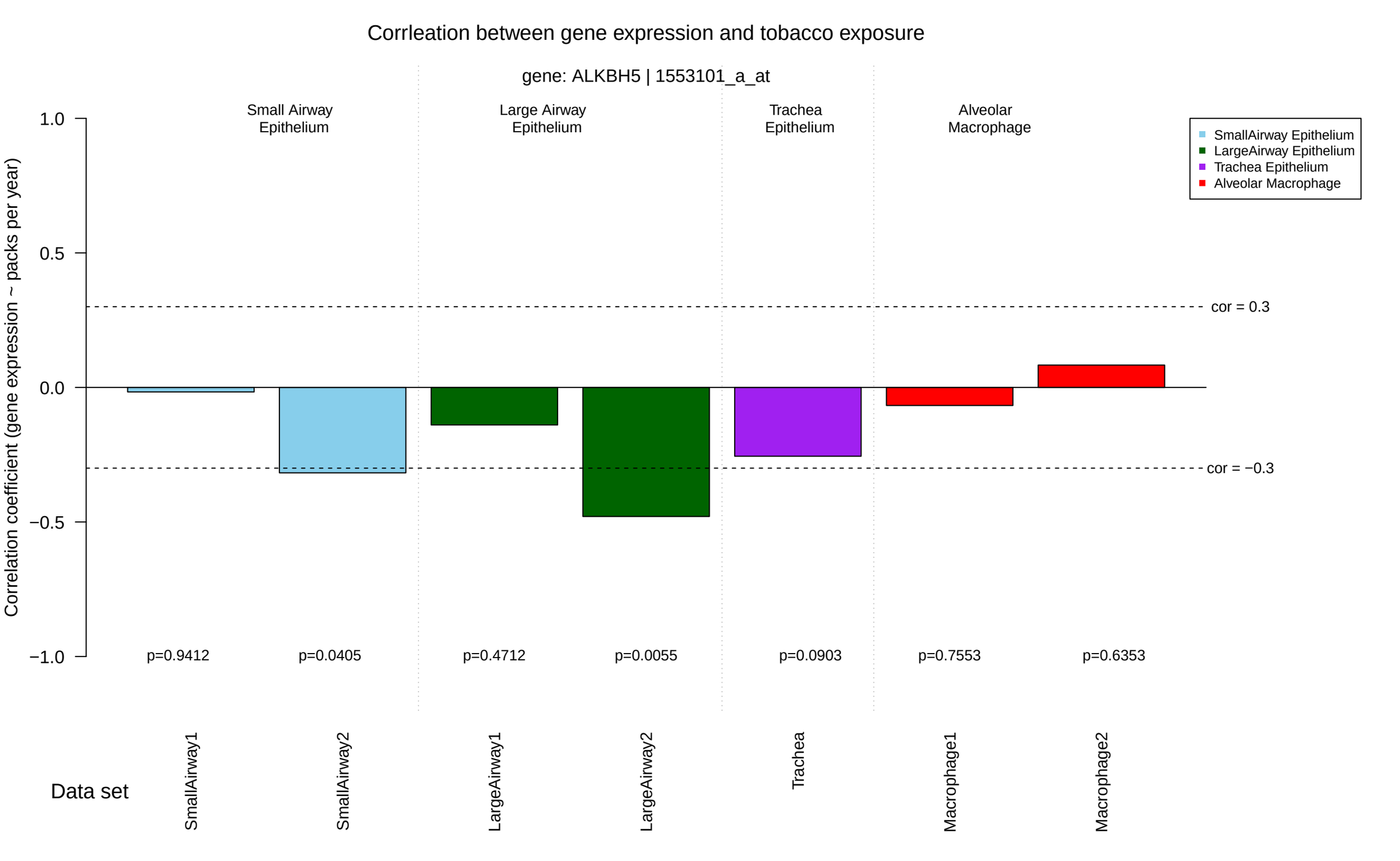

set: LargeAirway1 (p=0.1489)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0985)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0215)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8025)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1898)
ALKBH5 | 1553101_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH5 | 1553101_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9412
p=0.0405
p=0.4712
p=0.0055
p=0.0903
p=0.7553
p=0.6353
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2